

Temperature Fragility and the Conditional Benefits of Truncation Sampling

Anonymous ACL submission

Abstract

Large language models generate text by sampling each token from a predicted distribution, and a temperature parameter sets how far the draw strays from the most probable tokens. Truncation samplers such as top- p and min- p discard the least probable tokens before the draw, so that sampling at high temperature stays coherent. Their reported accuracy gains come from temperatures of 1.5 to 3, while deployed systems set defaults between 0.6 and 1.0. Whether they change accuracy at those defaults, and for which models, has not been measured. We test thirteen open-weight models on GSM8K and MMLU-Pro at temperatures 0.7, 1.0, and 1.3 in one controlled pipeline, seven of them under eight decoding configurations. Six of the thirteen models lose 17 to 38 accuracy points on MMLU-Pro between 0.7 and 1.3, and the other seven lose at most 10. The lost accuracy comes from generations that run to the token limit or never state an answer. These results suggest that truncation samplers improve accuracy primarily when higher temperatures substantially degrade model performance. Where accuracy remains stable across temperatures, none of the tested truncation samplers improves on plain temperature sampling.

1 Introduction

Sampling is the default way deployed language models generate text (Shi et al., 2024). At each step the model assigns a probability to every token in its vocabulary, and the decoder draws one from that distribution. A temperature parameter reshapes the distribution before the draw. Low temperatures concentrate probability on the few most likely tokens. High temperatures spread it toward the tail of the distribution, the many tokens the model rates as unlikely, where its estimates are least reliable (Holtzman et al., 2020; Hewitt et al., 2022). Model vendors and inference engines set

defaults between 0.6 and 1.0,¹ and every engine implements truncation samplers beside them. Truncation samplers, among them top- p , min- p , and top- $n\sigma$, discard the tail of the distribution before the draw, each by its own criterion. They were designed to keep open-ended text coherent at high temperature (Garces Arias et al., 2025), and they are now also evaluated on tasks with one checkable answer, where the claim is accuracy: min- p reports gains on GSM8K (Nguyen et al., 2025), and top- $n\sigma$ reports accuracy held up to a temperature of 3.0 (Tang et al., 2024).

Both results rest on one model and one range of temperatures. Top- $n\sigma$ was evaluated on Llama-3-8B (Tang et al., 2024) and min- p on Mistral-7B (Nguyen et al., 2025), at temperatures of 1.5 to 3, above the defaults systems set. A reanalysis that searched the same number of hyperparameter settings for every rule found the gains of min- p indistinguishable from those of the other rules (Schaeffer et al., 2025). The question cannot be settled by comparing across papers, because benchmark scores move with the inference backend even under greedy decoding (Masoudian et al., 2026), and with the prompt, the parser, and the item set. Settling it requires one pipeline in which everything except the decoding configuration is held fixed. It also requires enough models to show whether the answer depends on the model.

Research question: Under what model and temperature conditions do truncation samplers improve accuracy over plain temperature sampling?

We test thirteen open-weight models from the Llama-3, Mistral, Qwen, Gemma, and OLMo families on GSM8K and MMLU-Pro at temperatures

¹generation_config.json of meta-llama/Llama-3.1-8B-Instruct sets 0.6; llama.cpp’s common/sampling.h sets 0.8; the OpenAI chat-completions and Anthropic messages API references set 1.0. All read July 2026.

0.7, 1.0, and 1.3, in one pipeline with fixed items, prompts, token budgets, and parser. Ten of the models also run under eight decoding configurations: greedy decoding, plain temperature sampling, and six truncation configurations. Section 3 gives the design. Two questions follow. The first is whether a model keeps its accuracy between 0.7 and 1.3 under plain sampling, and what the lost accuracy looks like when it does not. The second is whether any truncation rule improves on plain temperature sampling between 0.7 and 1.3.

We find that temperature sensitivity differs by model. Between 0.7 and 1.3, six of the thirteen models lose 17 to 38 accuracy points on MMLU-Pro under plain sampling, and the other seven lose at most 10. The loss takes the form of a collapse of the output. On the six fragile models, the share of generations that run to the token limit or never state an answer rises by 26 to 78 points, while the share of generations that state a wrong answer does not rise. The collapse is a property of the model. Hermes-3, a different post-training of the same base model as Llama-3.1-8B, loses half as much accuracy as Llama-3.1-8B does. Truncation gains follow the collapse. On the models that hold their accuracy, no truncation sampler improves on plain temperature sampling at 0.7 or 1.0. On the models that collapse, every truncation sampler improves accuracy at 1.3, and at temperatures up to 2.0 truncation delays or prevents the collapse. The gains reported for truncation samplers are therefore real, and they are recoveries from a collapse that only some models suffer and that begins above the temperatures systems use. This work makes three contributions:

- **A controlled comparison of truncation samplers at the temperatures systems use.** Across thirteen open-weight models on GSM8K and MMLU-Pro, no truncation sampler improves on plain temperature sampling at 0.7 or 1.0, and an upper bound states how large a gain the results leave possible.
- **Temperature fragility as a property of the model.** Six of the thirteen models collapse between 1.0 and 1.3 into generations that run to the token limit or never state an answer. The collapse replicates across inference engine and precision. A different post-training of the same base model halves it. A plain temperature test identifies the fragile models before any sampler is compared.

- **The condition under which truncation samplers help.** On the collapsing models, every truncation sampler improves accuracy at 1.3 and delays the collapse at temperatures up to 2.0. This locates the gains reported for truncation samplers above the temperatures systems use, which reconciles those reports with the absence of gains below.

2 Related work

Truncation samplers. A truncation sampler removes the least probable tokens from the distribution before each token is sampled. Top- k keeps the k most probable tokens (Fan et al., 2018). Top- p , or nucleus sampling, keeps the smallest set of tokens whose probabilities sum to p , so the set grows when the distribution is flat (Holtzman et al., 2020). Min- p keeps every token whose probability is at least a fixed fraction of the largest (Nguyen et al., 2025). Top- $n\sigma$ works on the logits, the model’s scores before they are converted to probabilities, and keeps the tokens within n standard deviations of the largest logit (Tang et al., 2024). Temperature rescales all logits equally, so top- $n\sigma$ keeps the same tokens at every temperature, and so does top- k , because scaling does not change the order of the tokens. Top- p and min- p decide from the probabilities themselves, which temperature changes, so for them the order of scaling and truncation matters. A sampler applied after temperature scaling sees the scaled distribution and keeps a different set of tokens from the same sampler applied before it. Our main configurations apply temperature first, and for top- p and min- p we also run the reverse order, which is the default in llama.cpp.

Temperature-invariant samplers. The parameter of top- p or min- p fixes how much of the distribution is kept, and temperature reshapes the distribution. The same value therefore keeps a different set of tokens at each temperature, and a value that works well at one temperature need not work well at another. Two samplers published in 2026 begin from this observation. p -less sampling (Tan et al., 2026) and Min- k (Ding et al., 2026) each decide which tokens to keep from the shape of the distribution at every step, so that temperature does not change which tokens are kept. Neither paper measures whether that matters at the temperatures systems use. The evaluation of Min- k begins at temperature 1.0, where top- k , top- p , and min- p score within a point of one another on GSM8K,

and the samplers separate only above temperature 2 (Ding et al., 2026).

Evaluations of decoding and temperature. Studies that compare decoding methods find that the best one depends on the model. A survey of decoding methods reports that the best method changes with the task and the model (Shi et al., 2024). The best configuration for open-ended generation varies by model (Garces Arias et al., 2025), and so does the gap between greedy decoding and sampling on reasoning benchmarks (Song et al., 2025). Studies of temperature either stop at 1.0 or use tasks with little accuracy to lose. Renze and Guven (2024) find no effect of temperature between 0.0 and 1.0 on multiple-choice problems. Grover (2026) concludes that instruction tuning is the main determinant of robustness to temperature, on synthetic tasks where the most robust models score 11 to 30% and with no truncation sampler tested. Fastowski et al. (2025) link the entropy of the token distribution to temperature sensitivity on factual questions, and Troshin et al. (2025) attribute high-temperature losses on mathematics to wrong tokens at a few critical positions.

Conditions of measurement. One recent study shapes the design of our comparison. Masoudian et al. (2026) run three instruction-tuned models through five inference frameworks on six benchmarks and find that the framework changes the scores even under greedy decoding, by amounts that depend on the model. A comparison across models and samplers must therefore hold the framework fixed, or the differences it reports may belong to the framework. We run every configuration through one engine, llama.cpp. To check whether the collapse we observe comes from the engine, we rerun the most fragile model through a second engine at full precision. Our models also run quantized. Prasad and Pal (2026) find that quantization does not change how much a model loses to temperature, and our grid at four quantization levels agrees (Appendix C).

3 Experimental design

Table 1 lists the six runs. All share the same fixed items, client-rendered official chat templates, token budgets, parser, and seed derivation, so an accuracy difference between two runs is attributable to the factor that differs, up to the sampling variation the bootstrap quantifies.

Decoding. Eight configurations: greedy decod-

ing, plain temperature sampling, and the four truncation rules of Section 2 at the setting each source paper uses (top- p 0.95 (Holtzman et al., 2020), top- k 40 (Radford et al., 2019), min- p 0.05 (Nguyen et al., 2025), top- $n\sigma$ 1.0 (Tang et al., 2024)), with temperature scaling applied before truncation, plus two ablations that apply top- p and min- p before temperature scaling (the llama.cpp default order). Temperatures 0.7 and 1.0 bracket the defaults that models and engines set (Section 1), and 1.3 is a stress point above them. Each request carries exactly one truncation parameter, and the engine applies only that rule (Appendix A).

Tasks and grading. GSM8K, grade-school word problems (Cobbe et al., 2021), with the zero-shot chain-of-thought prompt (reasoning requested without worked examples) of Meta’s Llama evaluation suite (Meta AI, 2024c) and numeric exact match; and MMLU-Pro, ten-option multiple choice (Wang et al., 2024), zero-shot chain of thought and letter exact match. The fixed MMLU-Pro items are the first 50 of the test split, which is sorted by category, so all 50 are business questions; a stratified subset of 50 fresh items from all 14 categories checks this choice on three models (Appendix C). Both tasks need a chain of reasoning ending in one exactly checkable answer, and MMLU-Pro’s reasoning runs about twice as long (median 398 against 212 completion tokens at 0.7). The token budgets are 512 on GSM8K and 1024 on MMLU-Pro; a generation that reaches its budget is *capped*. Parsers follow lm-eval-harness (Gao et al., 2023) and log one of three outcomes: *strict*, the pinned final-answer sentence is present (“The final answer is 26”; “The answer is (I)”); *flexible*, the sentence is absent and the parser falls back to the last number or the last parenthesized letter; *failed*, neither exists, graded incorrect. Appendix B shows one generation per outcome. The two parsers treat degenerate text differently. Such text usually contains a number, so on GSM8K it is graded as a wrong answer through the flexible path; on MMLU-Pro it fails to parse. The capped flag records the failure on both tasks.

Models. The main grid holds seven instruction-tuned models from five families: Llama-3.1-8B (Grattafiori et al., 2024), Mistral-7B-v0.3 (Jiang et al., 2023), Qwen2.5-7B (Yang et al., 2024), Qwen3-1.7B, 4B, and 8B (Yang et al., 2025) with thinking off, and Gemma-3-12B (Gemma Team, 2025), each at four GGUF quantization levels (Q8_0, Q6_K, Q4_K_M, Q3_K_M) from one

Table 1: The six runs, 232,700 graded generations in total. Every run uses both tasks with the same fixed 50 items each and $K=3$ repetitions of every sampled configuration ($K=1$ for greedy decoding), except the stratified subset, which uses 50 fresh MMLU-Pro items. “8” configurations are the eight of the Decoding paragraph; K is the number of repetitions per item. The main grid includes a BF16 GGUF run of Qwen3-4B beside its four quantized runs. Gen. counts graded generations.

Run	Models	Precision, engine	Configurations	T	Gen.	Where
Main grid	7 (5 families)	Q8, Q6, Q4, Q3; llama.cpp	8	0.7, 1.0, 1.3	185,600	§4.1–4.4
Temperature extension	6 more	Q8; llama.cpp	greedy, plain temp.	0.7, 1.0, 1.3	6,000	§4.1, 4.2
Flagged-model grid	3 of the 6	Q8; llama.cpp	8	0.7, 1.0, 1.3	19,200	§4.3, 4.4
Engine and release checks	Llama-3.2, Llama-3.1, Qwen3-4B	BF16, INT8; transformers. F16, Q8; llama.cpp	greedy, plain temp.	0.7, 1.0, 1.3	6,000	§4.1, App. A
Temperature ladder	Llama-3.1, Qwen2.5, Gemma-3	Q8; llama.cpp	4	1.3, 1.5, 1.7, 2.0	14,400	§4.4
Stratified subset	Llama-3.1, Qwen2.5, Gemma-3	Q8; llama.cpp	greedy, plain temp.	0.7, 1.0, 1.3	1,500	App. C

provider and one recipe. Every temperature result uses Q8_0. The sampler bound of Section 4.3 pools the four levels, and is also given at Q8_0 alone. The temperature extension adds six models at Q8_0: two further Llama-3 releases, Llama-3-8B (Meta AI, 2024a) and Llama-3.2-3B (Meta AI, 2024b); Hermes-3-8B (Teknium et al., 2024), a different post-training of the Llama-3.1-8B base; Qwen3.5-9B (Qwen Team, 2026) and Gemma-4-E4B (Google DeepMind, 2026), two 2026 releases; and OLMo-3-7B (Allen Institute for AI, 2025), a fully open model. The four-model panel uses a later llama.cpp release than the grid’s pinned build, which two of its checkpoints require; Llama-3.2-3B and Qwen3-4B rerun on that release as a bridge (Section 4.1).

Fragility flag. Before the extension ran, we fixed a flag: a plain-temperature accuracy drop of at least 15 points from 0.7 to 1.3, a capped-rate rise of at least 20, or a strict-parse fall of at least 20, on either task. A flagged model receives the full eight-configuration grid at Q8_0. Three of the four 2026 panel models tripped it (Hermes-3, Qwen3.5-9B, OLMo-3-7B). The two further Llama releases ran before the flag was written and trip it as well; they received no grid, and Llama-3.1-8B’s main-grid rows stand for that lineage. We call the six 2026 and Llama additions the temperature extension, and the four 2026 models the panel.

Engine and precision. Every grid result comes from llama.cpp on quantized weights, so the collapse could in principle belong to the engine or the conversion. The engine check reruns the temperature test through Hugging Face Transformers on the official bfloat16 weights (BF16, a 16-bit floating-point format) of Llama-3.2-3B, in 8-bit weights for Llama-3.1-8B (the most a 16 GB card holds), and in BF16 for Qwen3-4B as a robust control, plus Llama-3.2-3B on its unquantized F16 GGUF through llama.cpp.

Inference. Every interval is a 95% nonpara-

metric bootstrap clustered on the 50 items (questions are resampled with all their generations together; $B=2000$, 5000 for the sampler bound). Both arms of every comparison ran the same items, so the bootstrap is paired. Item accuracy is the mean over repetitions, and over quantization levels where levels are pooled. The sampler bound is a simultaneous 95% upper confidence bound on the largest gain in a stated family of paired contrasts, of the max- t form: the largest observed gain plus the 95th percentile of the largest bootstrap deviation, with the contrasts resampled jointly (Westfall and Young, 1993). A generalized estimating equation over the same records reproduces the sampler-contrast intervals to within 0.6 points (Appendix E). Hardware, the pinned builds, model hashes, and the anonymized code and records are in Appendix A.

4 Results

4.1 Temperature sensitivity differs across checkpoints

Figure 1 gives the plain-temperature accuracy drop from 0.7 to 1.3 for all thirteen models at Q8_0, and Appendix C the accuracies at every temperature. Six models lose 17 to 38 points on MMLU-Pro: Llama-3.1-8B, Llama-3.2-3B, Qwen3.5-9B, Hermes-3-8B, OLMo-3-7B, and Llama-3-8B. The sensitivity is graded within the six. Three have interval lower limits at or above 15 (Llama-3.1, Llama-3.2, Qwen3.5); the other three trip the flag on point estimates, with intervals that overlap Mistral’s. The other seven lose at most 10, with intervals that all include 4 or less; we call these seven robust, meaning unflagged in the tested range. On GSM8K, the shorter task, Llama-3.1 and Llama-3.2 lose 30 and 38 points as well, and the other four fragile models lose 4 to 12; for those four the flag fires on MMLU-Pro only. Llama-3.1’s decline has begun by temperature 1.0, where its MMLU-Pro accuracy is already 10 points below its 0.7 level.

Accuracy at 0.7 does not predict the drop: Mistral-7B at 31% on GSM8K and Qwen3-4B at 96% both hold, while Llama-3.1 at 77% collapses. Model age does not predict it either, since Mistral-7B-v0.3 predates Llama-3.1.

The collapse is a property of the checkpoint under fixed conditions, and it survives a change of engine. Through Hugging Face Transformers on its official BF16 weights, Llama-3.2-3B loses 40.0 points on GSM8K and 36.0 on MMLU-Pro (Q8_0 through llama.cpp: 38.0 and 36.7), with 61 and 92% of its generations capped at 1.3; the paired differences between the two engines’ drops are +2.0 [−8.7, 12.0] and −0.7 [−9.3, 8.7]. Llama-3.1-8B in 8-bit weights loses 43.3 and 42.7, and Qwen3-4B in BF16 moves by 1.3 and −0.7, as in its GGUF cells. These runs replicate the collapse. They do not show that the engines are equivalent: the intervals on the differences span about 20 points, and Llama-3.1’s GSM8K drop is 13.3 points larger in 8-bit than at Q8_0 [2.7, 23.3] (Appendix A). Llama-3.2-3B and Qwen3-4B rerun on the later llama.cpp release reproduce their pinned-build drops with differences that include zero, and the stratified MMLU-Pro subset preserves Llama’s much larger drop (Llama-3.1 +43.3 [31.3, 55.3]; Qwen2.5 +6.0; Gemma-3 +7.3).

Hermes-3-8B shares the Llama-3.1-8B base and shows roughly half of its MMLU-Pro drop (19.3 against 38.0), with the same rise in capped generations. The two checkpoints differ only in post-training, so post-training changed the size of the collapse; which part of it did, the design cannot say. The six fragile checkpoints span three lineages, Llama-derived, Qwen, and OLMo, but the three Meta releases share data and methods, so they are not three independent observations.

4.2 The loss is capped or unparseable generations

This subsection uses plain temperature sampling throughout. An accuracy drop can come from two failures, completed answers that are wrong or generations that never reach an answer, and the parser fields separate them. Figure 2 splits every MMLU-Pro generation into three exclusive outcomes: correct; capped or unparseable (the generation reached its token budget, or contained no answer to grade); and answered wrong (a parsed answer that is wrong and ended before the budget). In all six fragile models the capped-or-unparseable share rises between 0.7 and 1.3, by 26 to 78 points, and the answered-

wrong share falls or stays flat (Llama-3.1: 31 to 3; Hermes-3: 49 to 29; Qwen3.5: 9 to 9). In the seven robust models the capped share moves by at most 11 points. GSM8K shows the same pattern for the two Llama checkpoints (Appendix B).

What the capped generations contain differs by temperature. A blinded audit of 152 sampled outputs, labelled by a language-model annotator that saw only the task and the raw text, found all eight audited Llama-3.1 generations capped at 1.3 to be incoherent token noise, all eight capped at 0.7 to be coherent reasoning cut off by the budget, and 15 of the 26 parse failures to contain a clear answer in an unexpected form; 34 of the 152 labels were marked borderline, and the audit covers four grid models and none of the flagged panel models (Appendix B). Budget censoring is visible at 0.7 in the verbose models: OLMo-3 and Qwen3.5 already cap 47 and 29% of their MMLU-Pro generations there. The audit establishes degeneration in the sampled Llama outputs at 1.3; the category as a whole also contains budget and formatting failures. Qwen3.5 is unaudited, and its parser fields show a rise of only 13 points in the capped rate against a fall of 31 in the strict-parse rate, so most of its generations that reach no answer end before the budget.

Among the generations that do complete with the pinned final-answer sentence, accuracy at 1.3 is subject to selection, because the surviving generations can come from a different mix of items. The matched comparison is greedy accuracy on the surviving generations’ items, each survivor counted once. On MMLU-Pro, Llama-3.1’s completed answers score 78.9% against 78.9% for matched greedy, Qwen3.5’s 83.1 against 84.7, and OLMo-3’s 73.3 against 73.3; for Llama-3-8B, Llama-3.2-3B (nine survivors), and Hermes-3 the completed answers score below matched greedy, by 4, 11, and 16 points (Appendix B). Three fragile checkpoints score 4 to 16 points below matched greedy among their surviving answers, in point estimates, and so does robust Mistral-7B (21 points), so a survivor gap at 1.3 is not specific to fragility (Table 5 gives the 0.7 reference for every model).

4.3 Measured sampler gains are small where plain temperature holds

Sampler gains are small on the six robust grid models at temperatures 0.7 and 1.0. For each model, task, and temperature we compare each truncation rule with plain temperature on the same items (all

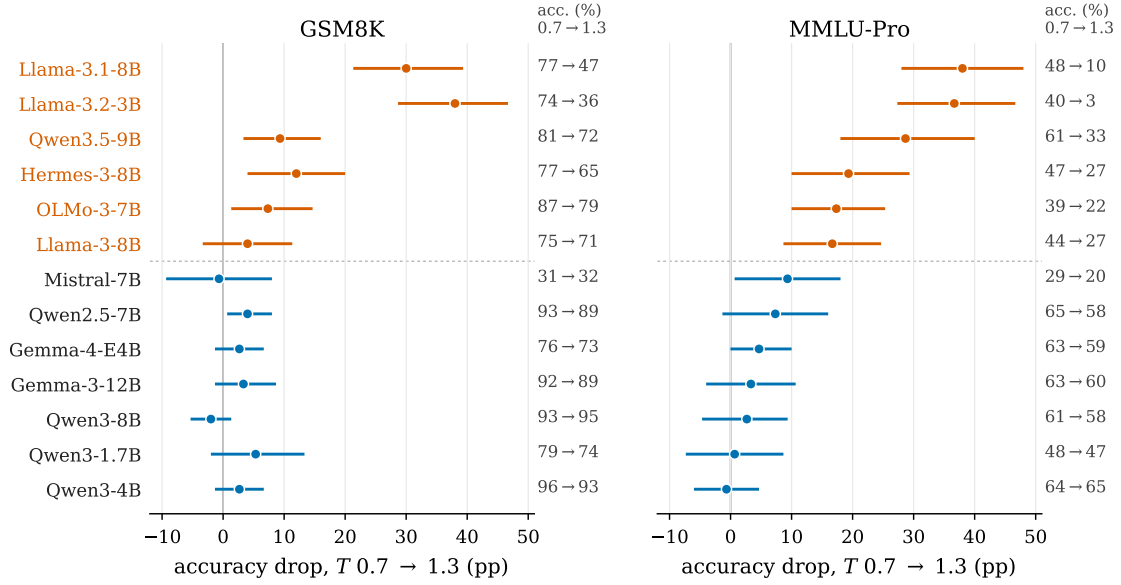


Figure 1: Accuracy drop under plain temperature sampling from $T=0.7$ to 1.3 at Q8_0, with 95% item-clustered bootstrap intervals; the accuracies at the two temperatures are printed at the right. Positive values are losses. Models are sorted by the MMLU-Pro drop; the six in orange above the dotted line lose at least 15 points on one task, the accuracy criterion of the fragility flag.

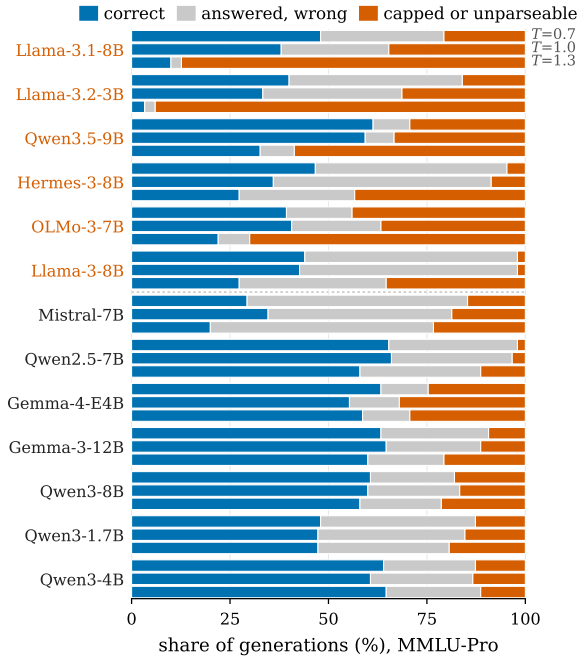


Figure 2: Outcome of every MMLU-Pro generation under plain temperature sampling at Q8_0, three bars per model, top to bottom $T=0.7$, 1.0, and 1.3. Categories are exclusive, in this precedence: correct; capped or unparseable (so this category holds incorrect generations only); answered wrong. Model order as in Figure 1.

largest gain over 96 comparisons is 3.3 points, 91 of the 96 point estimates lie within 3 points of zero, 1 interval lies above zero against 3 below, and the simultaneous 95% upper bound on the largest gain is 9.0. Restricted to Q8_0, the same 96 contrasts have a largest gain of 4.7, no interval above zero, and a bound of 16.0.

On the same six models at 1.3, gains reach 9 points and 13 of 48 intervals lie above zero, on Mistral, Qwen2.5, and Gemma-3, the three with the largest drops among the robust seven. Llama-3.1 is different already at 1.0: on MMLU-Pro, where its plain-temperature accuracy has fallen 10 points, min- p and top- $n\sigma$ gain 11 points (pooled), and on the flagged models' Q8_0 grids Hermes-3 gains 8 with top- $n\sigma$ at the same temperature.

The two ablations are outside these families. Applying top- p or min- p before temperature scaling changes accuracy by at most 4 points on the seven main-grid models at $T \leq 1.0$. The largest gain over plain temperature anywhere at $T \leq 1.0$ is the temperature-last top- p on Hermes-3 at 0.7 on MMLU-Pro, +12.0 [4.7, 19.3], one contrast among 48 ablation contrasts on the flagged models that otherwise stay within 7 points. On Llama-3.1 at 1.3, truncating first keeps 22 more points under top- p on MMLU-Pro: at that temperature the rule cuts a sharper distribution and removes more of the tail (Appendix E).

A bound of 9.0 points is an upper limit on what

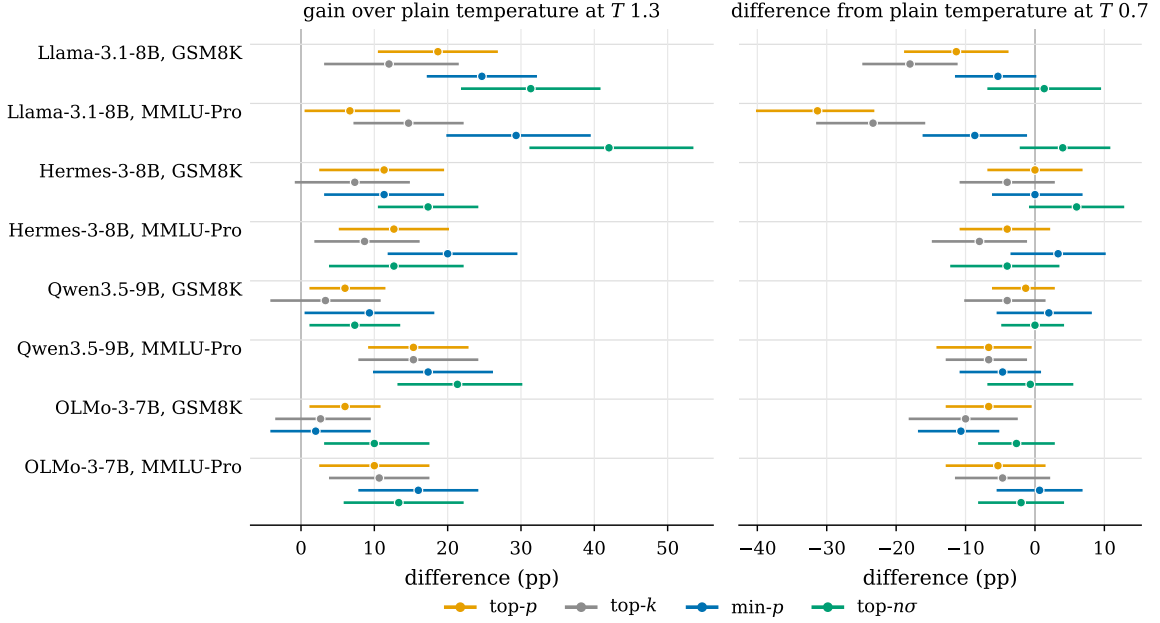


Figure 3: Truncation at $T=1.3$ on the four fragile checkpoints with a sampler grid (Llama-3.1-8B at Q8_0 from the main grid; the three flagged panel models). Rules at their standard settings with temperature applied first. Left: paired difference from plain temperature at 1.3. Right: paired difference from plain temperature at 0.7. Pointwise 95% item-clustered bootstrap intervals. For the flagged models both plain-temperature baselines come from the sampler-grid run.

Table 2: Paired gains of the four truncation rules over plain temperature, by family of contrasts: the number of contrasts, the largest point gain, the approximate 95% upper bound on the largest gain (the 95th percentile of the jointly resampled maximum; not computed for the families at 1.3, where the gains are the finding), and the number of pointwise intervals above zero. The six robust grid models are Mistral-7B, Qwen2.5-7B, Qwen3-1.7B, Qwen3-4B, Qwen3-8B, and Gemma-3-12B. Grid rows pool quantization levels; panel rows are Q8_0.

Models	T	contrasts	largest gain (pp)	95% bound (pp)	intervals > 0
six robust grid models	0.7, 1.0	96	+3.3	9.0	1
six robust grid models, Q8_0 only	0.7, 1.0	96	+4.7	16.0	0
six robust grid models	1.3	48	+9.0	—	13
Llama-3.1-8B	0.7, 1.0	16	+11.2	16.8	6
three flagged panel models	0.7, 1.0	48	+8.0	18.0	2
three flagged panel models	1.3	24	+21.3	32.7	20

the four rules can buy at their standard settings on these models and tasks. It is not evidence that the effect is zero, and it does not cover the fragile models, thinking mode, or open-ended generation.

4.4 Truncation recovers accuracy as plain temperature loses it

Across the 60 model, task, and temperature combinations with a sampler grid (seven grid models at Q8_0 and three flagged models), larger losses of plain temperature against greedy decoding accompany larger gains of truncation over plain tempera-

ture (Pearson $r=0.89$ over the 60 best-rule points; 0.78 over the 240 rule-level points). The association is descriptive, and the two quantities share the plain-temperature baseline (Appendix E). Figure 3 shows the recovery contrasts for the four fragile checkpoints with a grid at temperature 1.3. For the flagged models these contrasts use the plain-temperature baselines rerun within the sampler grid, which are a second run of the same cells with different seeds and differ from the extension run’s by up to 6 points (Hermes-3’s GSM8K accuracy at 0.7 is 77.3 there and 71.3 here; Qwen3.5’s MMLU-Pro accuracy at 1.3 is 32.7 and 38.0). Every rule’s point estimate is positive on every model and task, and 20 of the 24 intervals on the flagged models exclude zero. The right panel compares each rule with plain temperature at 0.7. In all eight model-task cells the best rule finishes within six points of that level, in point estimates: top- $n\sigma$ returns Llama-3.1’s MMLU-Pro accuracy from 10.0 to 52.0% and Qwen3.5’s from 38.0 to 59.3%, and min- p returns Hermes-3’s from 26.0 to 46.0% and OLMo-3’s from 26.7 to 42.7%. The recovery is not uniform: top- k leaves Hermes-3 8 points below its 0.7 level on MMLU-Pro, min- p leaves OLMo-3 11 points below on GSM8K, and top- p leaves Llama-3.1 31 points below on MMLU-Pro. Every rule was com-

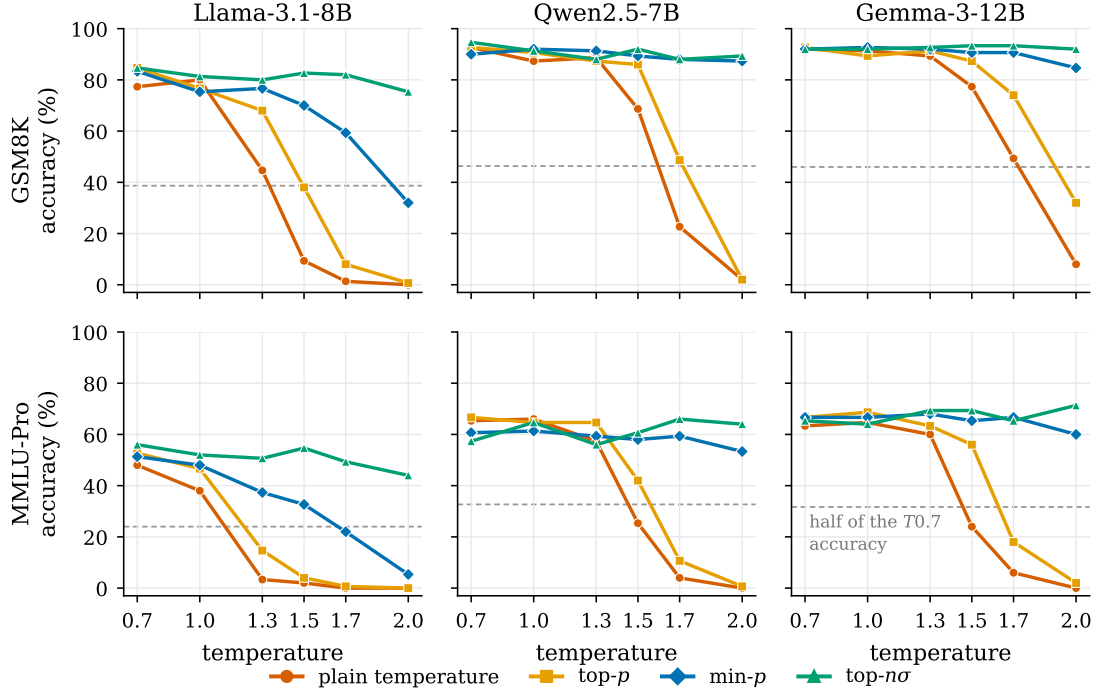


Figure 4: Accuracy against temperature under four configurations at Q8_0 (points at 0.7 and 1.0 from the main grid, 1.3 to 2.0 from the ladder run). The dashed line marks half of the same model and task’s plain-temperature accuracy at 0.7 in the main grid, the collapse criterion; crossings are read from point estimates on the tested temperatures.

pared with plain temperature only, and we do not rank the rules.

Where each model collapses. To see whether the robust models also collapse, the ladder run extends the temperature to 2.0 on Llama-3.1-8B, Qwen2.5-7B, and Gemma-3-12B under four configurations (Figure 4; drops with intervals in Appendix D). Call a configuration collapsed at the lowest tested temperature at which its accuracy falls below half of the same model and task’s plain-temperature accuracy at 0.7 (point estimates on the tested grid). Under plain temperature the three models collapse on GSM8K at 1.5, 1.7, and 2.0, in the order of Figure 1, and at 1.5 Llama’s drop interval is disjoint from the other two models’. On MMLU-Pro, the longer task, the crossings come at 1.3, 1.5, and 1.5. Truncation delays or avoids the crossing: top- p delays it in three of the six model-task combinations and leaves it unchanged in the other three; min- p delays it to 2.0 on Llama’s GSM8K and 1.7 on Llama’s MMLU-Pro and avoids it in the tested range on the other two models; top- $n\sigma$ never crosses it on any model or task through 2.0. Min- p and top- $n\sigma$ were validated at temperatures of 1.5 to 3 (Nguyen et al., 2025; Tang et al., 2024), a range that overlaps the collapse temperatures we observe, and that is where their

gains are largest here too.

5 Discussion and conclusion

Under plain temperature sampling, some checkpoints keep their accuracy above the default temperatures and others lose it by 1.3, and the loss appears as capped or unparseable generations. Larger plain-temperature losses accompany larger truncation gains; the recovery contrasts of Figure 3 carry that claim on their own, and the association across all cells is descriptive. This reconciles the sampler literature with the small gains we measure at default temperatures. Min- p and top- $n\sigma$ were evaluated in a range that overlaps the collapse temperatures we observe, and there they help; Min- k ’s comparison at temperature 1.0 shows the probability-based rules within a point of each other (Ding et al., 2026).

For a practitioner running objective tasks, the two settings should be chosen together. On a model that holds its accuracy at the intended temperature, the gain from a standard truncation rule is small: across 96 contrasts on the six robust grid models at 0.7 and 1.0, the largest estimated gain is 3.3 points and the simultaneous upper bound 9.0 (16.0 at Q8_0 alone). On a model that loses accuracy there, either a lower temperature or a truncation

rule recovers most of the loss, and the fragility can be measured without gold answers: the capped rate and the strict-parse rate need no labels, and their change between 0.7 and 1.3 separates the fragile models from the robust ones in every run of this study, including the panel run after the flag was written. We measure accuracy only, so we say nothing about the output diversity a higher temperature buys.

For sampler evaluations, a gain should be reported next to the plain-temperature loss it recovers, since a gain measured beyond one model’s collapse temperature can be absent on a model that collapses later. What moves the collapse temperature remains open. Hermes-3 shows that post-training changes its size on one base, and the grid contains no base-versus-instruct or checkpoint axis, so the cause is not identified here.

Limitations

Each task has 50 fixed items with $K=3$ repetitions; the small item count is what makes a factorial run of 185,600 generations feasible, and every main estimate carries an item-clustered interval. The fixed MMLU-Pro subset is single-category (business) because the test split is sorted by category; the stratified subset preserves Llama’s much larger drop on 50 fresh items, and as a second sample of 50 it does not estimate the benchmark-wide effect either. Mistral-7B scores 31% on GSM8K and 29% on MMLU-Pro at 0.7, which limits the power to detect a further drop on a model we count as robust.

Temperature 1.3 is a stress point above the defaults, chosen to bound the deployment range from above. The token budgets are fixed, and budget censoring is visible at 0.7 on the verbose models; the audit distinguishes cut-off reasoning from degenerate text on a sample of 152 outputs from four grid models, labelled by a language model with 34 borderline cases; the remaining records, including every output of the three flagged panel models, are unaudited. The survivor comparison controls for item composition with matched greedy decoding; it does not remove the selection.

The sampler evidence is narrower than the temperature evidence. Thirteen models ran the temperature test; ten ran a sampler grid, and the three flagged models were selected by their temperature drop, so the recovery estimates describe a fragility-defined subgroup. “Best rule” is chosen after the fact, which favours the recovery. Rules were tested

at one setting each, in one engine, and the bound applies to those settings. The two engine checks replicate the collapse for two Llama checkpoints and one control; Llama-3.1’s check ran in 8-bit weights, its native precision untested, and the intervals on the between-engine differences are about 20 points wide. Repeated generations, temperatures, and rules are not independent observations, so the gain-versus-loss correlation is descriptive.

The scope is open-weight instruction-tuned models of 1.7B to 12B parameters, in English, zero-shot, on objective tasks with one checkable answer, with thinking mode off. A spot-check of Qwen3-8B with thinking on shows a 10-point MMLU-Pro drop [2, 20] at $N=25$ (Appendix C), so the bound does not cover thinking mode. Open-ended generation, the original use of truncation samplers, lies outside the design, as does the diversity a higher temperature buys under multi-sample selection. The study identifies no cause for the differences between checkpoints.

References

- Allen Institute for AI. 2025. OLMo 3 7b instruct model card. <https://huggingface.co/allenai/olmo-3-7B-Instruct>.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*.
- Yuanhao Ding, Meimingwei Li, Esteban Garces Arias, Matthias Aßenmacher, Christian Heumann, and Chongsheng Zhang. 2026. Min- k sampling: Decoupling truncation from temperature scaling via relative logit dynamics. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14932–14948.
- Angela Fan, Mike Lewis, and Yann Dauphin. 2018. Hierarchical neural story generation. In *Proceedings of ACL*.
- Alina Fastowski, Bardh Prenkaj, and Gjergji Kasneci. 2025. From confidence to collapse in LLM factual robustness. *arXiv preprint arXiv:2508.16267*.
- Leo Gao, Jonathan Tow, Baber Abbasi, Stella Biderman, Sid Black, et al. 2023. A framework for few-shot language model evaluation. Zenodo.
- Esteban Garces Arias, Meimingwei Li, Christian Heumann, and Matthias Aßenmacher. 2025. Decoding decoded: Understanding hyperparameter effects

689	in open-ended text generation. In <i>Proceedings of the 31st International Conference on Computational Linguistics (COLING)</i> .	Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. Language models are unsupervised multitask learners . <i>OpenAI technical report</i> .	741
690			742
691			743
692	Gemma Team. 2025. Gemma 3 technical report . <i>arXiv preprint arXiv:2503.19786</i> .	Matthew Renze and Erhan Guven. 2024. The effect of sampling temperature on problem solving in large language models . In <i>Findings of EMNLP</i> .	745
693			746
694	Google DeepMind. 2026. Gemma 4 E4B instruction-tuned model card. https://huggingface.co/google/gemma-4-E4B-it .		747
695			
696			
697	Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. 2024. The Llama 3 herd of models . <i>arXiv preprint arXiv:2407.21783</i> .	Rylan Schaeffer, Joshua Kazdan, and Yegor Denisov-Blanch. 2025. Min-p, max exaggeration: A critical analysis of min-p sampling in language models . <i>arXiv preprint arXiv:2506.13681</i> .	748
698			749
699			750
700	Kabir Grover. 2026. Reliability under randomness: An empirical analysis of sparse and dense language model behavior across decoding temperatures . <i>arXiv preprint arXiv:2601.00942</i> .		751
701			
702			
703			
704	John Hewitt, Christopher D. Manning, and Percy Liang. 2022. Truncation sampling as language model desmoothing . In <i>Findings of EMNLP</i> .	Chufan Shi, Haoran Yang, Deng Cai, Zhisong Zhang, Yifan Wang, Yujiu Yang, and Wai Lam. 2024. A thorough examination of decoding methods in the era of LLMs . In <i>Proceedings of EMNLP</i> .	752
705			753
706			754
707			755
708	Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. 2020. The curious case of neural text de-generation . In <i>International Conference on Learning Representations (ICLR)</i> .	Yifan Song, Guoyin Wang, Sujian Li, and Bill Yuchen Lin. 2025. The good, the bad, and the greedy: Evaluation of LLMs should not ignore non-determinism . In <i>Proceedings of NAACL</i> .	756
709			757
710			758
711			759
712	Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. 2023. Mistral 7B . <i>arXiv preprint arXiv:2310.06825</i> .	Runyan Tan, Shuang Wu, and Phillip Howard. 2026. p-less sampling: A robust hyperparameter-free approach for LLM decoding . In <i>International Conference on Learning Representations (ICLR)</i> .	760
713			761
714			762
715			763
716	Shahed Masoudian, Passant Shafaei, Monorama Swain, and Markus Schedl. 2026. What we observe as LLM behavior can be a side-effect of inference backend . <i>arXiv preprint arXiv:2608.04714</i> .	Chenxia Tang, Jianchun Liu, Hongli Xu, and Liusheng Huang. 2024. Top-$n\sigma$: Not all logits are you need . <i>arXiv preprint arXiv:2411.07641</i> .	764
717			765
718			766
719			
720	Meta AI. 2024a. Introducing Meta Llama 3: The most capable openly available LLM to date. https://ai.meta.com/blog/meta-llama-3/ .	Ryan Teknium, Jeffrey Quesnelle, and Chen Guang. 2024. Hermes 3 technical report . https://arxiv.org/abs/2408.11857 . <i>Preprint</i> , arXiv:2408.11857.	767
721			768
722			769
723	Meta AI. 2024b. Llama 3.2: Revolutionizing edge AI and vision with open, customizable models. https://ai.meta.com/blog/llama-3-2-connect-2024-vision-edge-mobile-devices/ .	Sergey Troshin, Irina Mohammed, Yan Meng, Christof Monz, Antske Fokkens, and Vlad Niculae. 2025. Control the temperature: Selective sampling for diverse and high-quality LLM outputs . In <i>Conference on Language Modeling (COLM)</i> .	770
724			771
725			772
726			773
727	Meta AI. 2024c. Meta Llama 3 evaluation details. https://github.com/meta-llama/llama3/blob/main/eval_details.md .	Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhui Chen. 2024. MMLU-Pro: A more robust and challenging multi-task language understanding benchmark . In <i>Advances in Neural Information Processing Systems</i> .	774
728			775
729			776
730	Minh Nhat Nguyen, Andrew Baker, Clement Neo, Allen Roush, Andreas Kirsch, and Ravid Shwartz-Ziv. 2025. Turning up the heat: Min-p sampling for creative and coherent LLM outputs . In <i>International Conference on Learning Representations (ICLR)</i> .		777
731			778
732			779
733			780
734			781
735	Hari Prasad and Ritam Pal. 2026. The joint effect of quantization and sampling temperature on LLM safety alignment: A factorial analysis . <i>arXiv preprint arXiv:2606.29581</i> .	Sean Welleck, Ilia Kulikov, Jaedeok Kim, Richard Yuanzhe Pang, and Kyunghyun Cho. 2020. Consistency of a recurrent language model with respect to incomplete decoding . In <i>Proceedings of EMNLP</i> .	782
736			783
737			784
738			785
739	Qwen Team. 2026. Qwen3.5-9b model card. https://huggingface.co/Qwen/Qwen3.5-9B .	Peter H. Westfall and S. Stanley Young. 1993. <i>Resampling-Based Multiple Testing: Examples and Methods for p-Value Adjustment</i> . Wiley, New York.	786
740			787
		An Yang et al. 2024. Qwen2.5 technical report . <i>arXiv preprint arXiv:2412.15115</i> .	788
			789
			790
		An Yang et al. 2025. Qwen3 technical report . <i>arXiv preprint arXiv:2505.09388</i> .	791
			792
			793
			794

A Reproducibility and engine checks

Inference stack. llama.cpp at commit 5aba5364 (server build 9456), native /completion endpoint, CUDA builds per GPU architecture (sm_75 for an RTX 2080 Ti, sm_120 for an RTX 5070 Ti). Server-side BOS insertion is disabled; the client-rendered chat template owns the special tokens, and each record stores the SHA-256 of the rendered prompt. The 2026 panel, its sampler grids, and the bridge run use release v0.4.0 (commit 5266f24d), which two panel models require, with the same client-side rendering and the server’s template parser off. That release forces its own BOS for Gemma 4, so the Gemma-4 prompt is rendered without one; on both builds the server returns some cap-length degenerate streams with a zero token count, the pinned build with their text (which the client recovers, and which the audit labelled) and v0.4.0 as an HTTP error without text. The runner records those as zero-token capped generations. They number 189 of 185,600 records in the main grid, 114 in the Llama-releases run, 158 in the panel, 196 in its sampler grids, 64 in the engine check, 73 in the bridge, and 2,549 on the ladder; at $T=1.3$ under plain temperature they are 13% of Llama-3.1’s capped generations. Every quantized checkpoint is a Bartowski imatrix GGUF; SHA-256 hashes of the model files are in the repository (DOWNLOADS.md).

Decoding and seeding. Exactly one truncation parameter per request; the chain order is pinned through the server’s samplers array and recorded in every record. At both builds the server instantiates only the samplers named in that array, so its defaults for unlisted samplers (top- k 40, top- p 0.95, min- p 0.05) are inert, and no penalty, DRY, XTC, or logit-bias parameter is sent. Every generation’s seed is derived from (global seed 0, model, quantization, sampler, temperature, task, item, repetition). Batched serving (≤ 8 slots) is seed-logged but not bit-exact. Tasks: GSM8K (openai/gsm8k, test split, first 50 items) and MMLU-Pro (TIGER-Lab/MMLU-Pro, test split, first 50 items). Each generation is one JSONL record with the identity tuple, request parameters, raw output, parse outcome, and token counts; runs are resumable by record id. The pipeline, configurations, analysis scripts, and every record are at <https://anonymous.4open.science/r/decoding-robustness-027C>.

Engine check. The transformers configuration loads meta-llama/Llama-3.2-3B-Instruct

(revision 0cb88a4f) in BF16 with transformers 5.16.1 and torch 2.14, samples with do_sample at temperature T with top- k and top- p disabled and no other processor, in batches of eight with left padding; greedy decoding is do_sample=False. Generation stops on the model’s end-of-sequence set or at the task’s budget. The prompt is rendered from the same template as the grid; the template stamps the run date into the Llama system header, the only difference between the runs’ prompt strings, and the prompt token counts agree on every item. Llama-3.1-8B loads the unsloth/Llama-3.1-8B-Instruct mirror of the official weights (revision 4699cc75) in 8-bit through bitsandbytes, in batches of four; Qwen3-4B loads Qwen/Qwen3-4B (revision 1cfa9a72) in BF16 with thinking disabled. Both reproduce every grid prompt hash. Table 3 gives accuracy levels, drops, capped rates, and the paired difference of each configuration’s drop from the pinned Q8_0 llama.cpp cell, including the v0.4.0 bridge rows.

B Grading and degeneration

Figure 5 shows one generation per parse outcome and Figure 6 two degenerate outputs. The strict rule matches the pinned sentence with the answer directly after it. The flexible rule catches two different things: a well-formed answer written outside the pinned sentence, and an incoherent generation that still contains a number, which GSM8K grades as a wrong answer. Mistral omits the pinned sentence on 26 to 32% of its GSM8K generations at every temperature, and the Qwen3 models wrap the answer in markdown emphasis, which the strict rule does not match; both grade correctly through the flexible rule. Because markdown defeats the strict rule, we re-graded every main-grid record with a strict rule that tolerates it: 598 of 185,600 grades change, 556 of them on Qwen3-8B and Qwen3-4B MMLU-Pro, and no plain-temperature drop moves by more than 1.0 point. The original parser, fixed before any run, stays primary. Table 4 gives the share of each outcome and the capped rate for every model and temperature, and Figure 7 the GSM8K outcome composition.

Blinded audit. To check that the parser outcomes mean what Section 4.2 takes them to mean, we drew 152 plain-temperature generations at Q8_0 from Llama-3.1-8B, Mistral-7B, Qwen2.5-7B, and Qwen3-8B at $T \in \{0.7, 1.3\}$ on both tasks, stratified by parse outcome with the uncertain out-

Table 3: Engine, precision, and release checks: plain-temperature accuracy at $T=0.7$ and 1.3, the drop with its 95% item-clustered interval, the capped rate at 1.3 (%), and the difference of the drop from the reference Q8_0 llama.cpp cell on the same items (positive: a larger drop than the reference). The BF16 GGUF row of Qwen3-4B belongs to the main grid.

Model	Precision, engine	GSM8K					MMLU-Pro				
		0.7	1.3	drop [CI]	cap	vs. ref. [CI]	0.7	1.3	drop [CI]	cap	vs. ref. [CI]
Llama-3.2-3B	BF16, transformers	74.7	34.7	+40.0 [29.3, 49.3]	61	+2.0 [-8.7, 12.0]	40.7	4.7	+36.0 [24.7, 48.0]	92	-0.7 [-9.3, 8.7]
	F16 GGUF, llama.cpp	74.0	42.7	+31.3 [22.7, 40.7]	53	-6.7 [-17.3, 4.0]	38.0	5.3	+32.7 [23.3, 43.3]	91	-4.0 [-10.7, 2.7]
	Q8 GGUF, llama.cpp v0.4.0	77.3	34.7	+42.7 [34.0, 50.7]	63	+4.7 [-3.3, 12.7]	39.3	4.7	+34.7 [24.7, 45.3]	93	-2.0 [-9.3, 5.3]
	Q8 GGUF, llama.cpp (reference)	74.0	36.0	+38.0 [28.7, 46.7]	55	-	40.0	3.3	+36.7 [27.3, 46.7]	92	-
Llama-3.1-8B	INT8, transformers	82.7	39.3	+43.3 [33.3, 52.0]	54	+13.3 [2.7, 23.3]	45.3	2.7	+42.7 [32.7, 53.3]	95	+4.7 [-3.3, 12.7]
	Q8 GGUF, llama.cpp (reference)	77.3	47.3	+30.0 [21.3, 39.3]	51	-	48.0	10.0	+38.0 [28.0, 48.0]	86	-
Qwen3-4B	BF16, transformers	94.7	93.3	+1.3 [-1.3, 4.0]	0	-1.3 [-5.3, 2.0]	60.7	61.3	-0.7 [-6.0, 5.3]	13	+0.0 [-6.7, 6.7]
	BF16 GGUF, llama.cpp	95.3	96.7	-1.3 [-4.0, 1.3]	0	-4.0 [-9.3, 1.3]	58.0	60.0	-2.0 [-7.3, 3.3]	15	-1.3 [-8.7, 6.0]
	Q8 GGUF, llama.cpp v0.4.0	96.7	94.0	+2.7 [-0.7, 7.3]	0	+0.0 [-4.0, 4.7]	65.3	64.0	+1.3 [-4.7, 7.3]	11	+2.0 [-6.0, 10.0]
	Q8 GGUF, llama.cpp (reference)	96.0	93.3	+2.7 [-1.3, 6.7]	1	-	64.0	64.7	-0.7 [-6.0, 4.7]	12	-

[strict] ... $x = 19.50 / 0.75$ $x = 26$ So the original price of the book was \$26. The final answer is \$26.

[flexible] ... 80 \text{ miles} + 150 \text{ miles} = 230 \text{ miles} ### Final Answer: The final answer is 230.

[failed] ... $\frac{3,462.20}{35} = 98.92$ ### Final Answer: The answer is J.

Figure 5: One generation per parse outcome (endings only). Strict: Llama-3.1-8B, GSM8K, $T=1.3$, graded 26, correct. Flexible: Qwen3-8B, GSM8K, $T=0.7$; the emphasis markers defeat the strict rule and the last-number fallback grades 230, correct. Failed: Qwen3-8B, MMLU-Pro, $T=0.7$; the parser expects a parenthesized letter such as (J), none appears, graded incorrect.

[opens] To solve this problem, we'll use the gross profit method of inventory evaluation. Step 1: Calculate the gross profit earned during the year. . .
[ends] H sieve Con Palette sediment conduit France called coaching OMAX may PHYS today Enhanced vertices XOR/t and registration ordinances efforts yesterday Michael

[opens] To determine the correct type of research methods, we need to understand the characteristics of each option: A. Non-probability: This method involves selecting. . .
[ends] entageeba lever modeling ign measures inherits cones quantum auditingippines image later campaigns Taylorart parte diesel El N resembled Dental professionals

Figure 6: Two degenerate outputs (Llama-3.1-8B, Q8_0, plain temperature, $T=1.3$, MMLU-Pro; both reached the 1024-token budget). Each opens coherently, loses coherence mid-generation, and continues as token noise until the budget.

comes oversampled, and had them labelled by a language-model annotator (Claude) that saw only the task and the raw output: completed, cut off

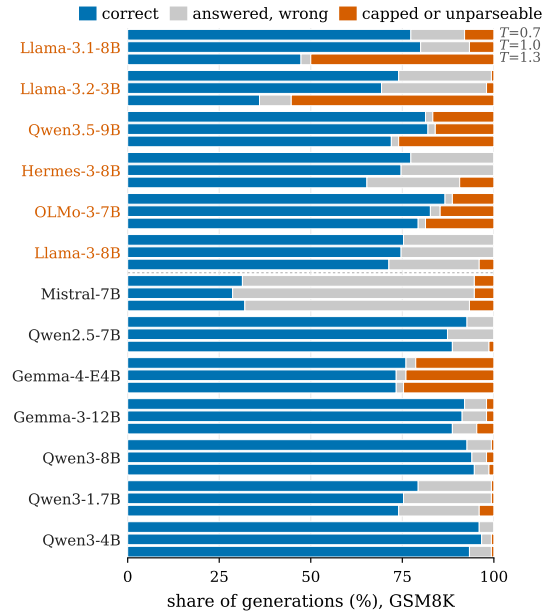


Figure 7: Outcome of every GSM8K generation under plain temperature sampling at Q8_0, as in Figure 2. On GSM8K a degenerate generation usually contains a number, so part of the degeneration is graded through the flexible path; the capped flag records it.

while coherent, degenerate, misformatted, or other. Of the 51 capped outputs, all 8 of Llama's at 1.3 are degenerate and all 8 at 0.7 are coherent reasoning cut short; the robust models' 21 capped outputs at 1.3 split 8 degenerate, 9 cut off, 3 completed, and 1 misformatted, and their 14 at 0.7 split 13 cut off and 1 completed. Of the 26 parse failures, 15 contain a clear answer in an unexpected form, 6 are degenerate, 4 are refusals, and 1 is cut off. Of the 75 strict and flexible parses, 72 are completed answers. The annotator marked 34 of the 152 labels as borderline. Sample, labels, and key are in the repository (results/outcome_audit).

Survivor comparison. Table 5 gives, for ev-

ery model, the share of plain-temperature generations with a strict parse at 0.7 and 1.3, the accuracy among them, and greedy accuracy on the same surviving items.

C Temperature results in full

Table 6 gives plain-temperature accuracy at all three temperatures for the thirteen models, the drop with its interval, and the change in capped and strict-parse rates. Table 7 recomputes the drop at each quantization level of the main grid: Llama-3.1 collapses at every level and no robust model exceeds 12 points at any level. Quantization itself costs little greedy accuracy: from Q8_0 to Q3_K_M, greedy accuracy pooled over both tasks changes by at most 5 points on the models of 7B parameters or more and falls by 7 and 9 points on Qwen3-4B and Qwen3-1.7B. Table 8 repeats the temperature test on a stratified MMLU-Pro subset with 3 to 4 items from each of the 14 categories; Llama’s drop stays far above the other two, and Gemma-3’s stratified drop, +7.3 [2.0, 12.7], has an interval that excludes zero. A spot-check of Qwen3-8B with thinking on (Q8_0, $N=25$, $K=2$, 450 generations) reproduces the GSM8K result (94% at 0.7 against 96% at 1.3) and not the MMLU-Pro one (56 against 46%, a 10-point drop with interval [2, 20]); with thinking on, the longer traces make Qwen3-8B less robust on the longer task.

D Temperature ladder

Table 9 gives the plain-temperature drop at each ladder temperature against the model’s own accuracy at 0.7, with intervals. The ladder is a separate run from the main grid, and its 1.3 step differs from Table 6 by amounts within the intervals (Llama-3.1 on MMLU-Pro: +44.7 against +38.0).

E Sampler results in full

Tables 10 and 11 give the accuracy of every configuration at every temperature for the ten models with a sampler grid (main-grid models pooled over quantization levels; flagged models at Q8_0). Tables 12 and 13 give every paired contrast against plain temperature, including the two temperature-last ablations; an interval endpoint within 10^{-9} of zero counts as touching zero. A generalized estimating equation over the same records (identity link, item clusters, exchangeable working correlation) gives the same point estimates and intervals

that differ by at most 0.6 points (median 0.1). Figure 8 plots the gain of each rule against the loss of plain temperature relative to greedy decoding, 240 points over 60 combinations. The two quantities share the plain-temperature baseline, so the association is descriptive.

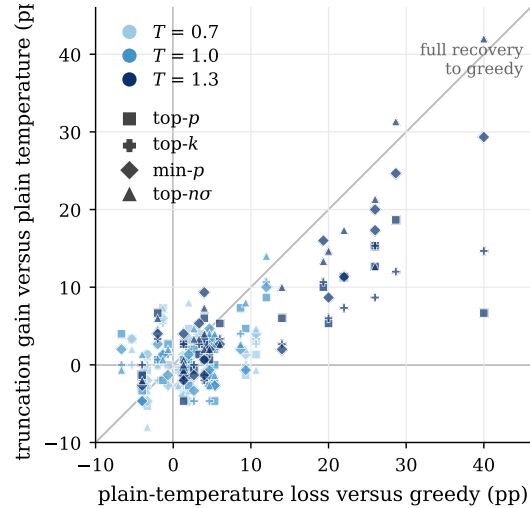


Figure 8: Gain of each truncation rule over plain temperature against the loss of plain temperature relative to greedy decoding: 240 points, four rules in each of the 60 model, task, and temperature combinations with a sampler grid (Q8_0). Both axes subtract the same plain-temperature accuracy. The diagonal marks full recovery to greedy accuracy.

Table 4: Parse outcomes under plain temperature at Q8_0 (% of generations, rounded): strict, flexible, failed, and capped (capped overlaps the other three).

Model	T	GSM8K				MMLU-Pro			
		strict	flex.	failed	capped	strict	flex.	failed	capped
Llama-3.1-8B	0.7	89	11	0	9	79	1	19	11
	1.0	87	13	0	7	63	5	32	22
	1.3	45	55	0	51	13	6	81	86
Llama-3.2-3B	0.7	99	1	0	1	83	3	15	10
	1.0	89	11	0	2	65	5	30	22
	1.3	41	59	0	55	6	9	85	92
Qwen3.5-9B	0.7	80	20	0	21	71	3	26	29
	1.0	79	21	0	21	67	5	27	33
	1.3	71	29	0	27	39	9	51	43
Hermes-3-8B	0.7	68	32	0	0	93	2	5	0
	1.0	68	32	0	0	91	1	9	0
	1.3	59	39	2	9	57	2	41	38
OLMo-3-7B	0.7	84	16	0	14	56	11	33	47
	1.0	84	16	0	16	62	10	28	38
	1.3	78	17	5	22	30	0	70	68
Llama-3-8B	0.7	98	2	0	0	97	1	2	0
	1.0	98	2	0	0	97	1	2	0
	1.3	94	6	0	5	65	4	31	29
Mistral-7B	0.7	72	28	0	5	82	4	14	1
	1.0	73	27	0	6	81	2	17	1
	1.3	71	29	0	7	74	6	20	11
Qwen2.5-7B	0.7	97	3	0	0	98	1	1	2
	1.0	93	7	0	0	96	1	3	3
	1.3	93	7	0	1	88	3	9	9
Gemma-4-E4B	0.7	45	55	0	25	74	8	18	26
	1.0	54	46	0	28	67	8	25	34
	1.3	49	51	0	29	68	10	22	32
Gemma-3-12B	0.7	97	3	0	3	91	1	9	9
	1.0	98	2	0	2	89	3	9	11
	1.3	95	5	0	5	79	3	17	20
Qwen3-8B	0.7	45	55	0	1	82	1	17	12
	1.0	45	55	0	2	83	5	13	13
	1.3	47	53	0	2	78	5	17	18
Qwen3-1.7B	0.7	15	85	0	1	87	2	11	13
	1.0	22	78	0	1	85	1	14	14
	1.3	21	79	0	4	83	2	15	19
Qwen3-4B	0.7	96	4	0	1	89	0	11	10
	1.0	93	7	0	1	87	1	12	12
	1.3	96	4	0	1	88	1	11	12

Table 5: Accuracy among strict-parsed generations under plain temperature at Q8_0, with the share of generations that parse strictly, the number n of such generations at $T=1.3$ (of 150), and greedy accuracy on their items with each surviving generation counted once. Point estimates without intervals.

Model	Task	$T=0.7$			$T=1.3$			
		strict %	acc strict	greedy, same items	strict %	n	acc strict	greedy, same items
Llama-3.1-8B	GSM8K	89	85.1	81.3	45	67	98.5	95.5
	MMLU-Pro	79	60.5	56.3	13	19	78.9	78.9
Llama-3.2-3B	GSM8K	99	74.3	74.3	41	62	85.5	90.3
	MMLU-Pro	83	47.6	48.4	6	9	55.6	66.7
Qwen3.5-9B	GSM8K	80	97.5	92.5	71	106	99.1	93.4
	MMLU-Pro	71	86.8	81.1	39	59	83.1	84.7
Hermes-3-8B	GSM8K	68	86.3	83.3	59	89	83.1	82.0
	MMLU-Pro	93	50.0	54.3	57	86	47.7	64.0
OLMo-3-7B	GSM8K	84	98.4	97.6	78	117	97.4	95.7
	MMLU-Pro	56	69.0	61.9	30	45	73.3	73.3
Llama-3-8B	GSM8K	98	76.9	76.9	94	141	75.2	77.3
	MMLU-Pro	97	45.2	39.7	65	98	41.8	45.9
Mistral-7B	GSM8K	72	33.3	26.9	71	106	32.1	26.4
	MMLU-Pro	82	35.0	39.8	74	111	25.2	45.9
Qwen2.5-7B	GSM8K	97	94.5	93.8	93	139	91.4	93.5
	MMLU-Pro	98	66.7	63.3	88	132	65.2	68.2
Gemma-4-E4B	GSM8K	45	98.5	94.0	49	74	93.2	90.5
	MMLU-Pro	74	83.8	81.1	68	102	82.4	83.3
Gemma-3-12B	GSM8K	97	93.8	90.4	95	143	93.0	93.0
	MMLU-Pro	91	69.9	62.5	79	119	75.6	67.2
Qwen3-8B	GSM8K	45	97.0	98.5	47	70	100.0	100.0
	MMLU-Pro	82	74.0	70.7	78	117	73.5	74.4
Qwen3-1.7B	GSM8K	15	91.3	91.3	21	31	71.0	80.6
	MMLU-Pro	87	55.0	55.0	83	124	57.3	54.8
Qwen3-4B	GSM8K	96	95.8	97.9	96	144	93.8	97.9
	MMLU-Pro	89	72.2	72.2	88	132	73.5	73.5

Table 6: Plain-temperature accuracy at Q8_0 for all thirteen models, the drop from $T=0.7$ to 1.3 with its 95% item-clustered interval (bold: lower limit at or above 15, a stricter criterion than the fragility flag), and the change in capped and strict-parse rates. Models sorted as in Figure 1.

Model	Task	accuracy (%) at T			drop [95% CI]	change 0.7→1.3 (pp)	
		0.7	1.0	1.3		capped	strict
Llama-3.1-8B	GSM8K	77.3	80.0	47.3	+ 30.0 [21.3, 39.3]	+43	-45
	MMLU-Pro	48.0	38.0	10.0	+ 38.0 [28.0, 48.0]	+75	-67
Llama-3.2-3B	GSM8K	74.0	69.3	36.0	+ 38.0 [28.7, 46.7]	+55	-57
	MMLU-Pro	40.0	33.3	3.3	+ 36.7 [27.3, 46.7]	+82	-77
Qwen3.5-9B	GSM8K	81.3	82.0	72.0	+9.3 [3.3, 16.0]	+6	-9
	MMLU-Pro	61.3	59.3	32.7	+ 28.7 [18.0, 40.0]	+13	-31
Hermes-3-8B	GSM8K	77.3	74.7	65.3	+12.0 [4.0, 20.0]	+9	-9
	MMLU-Pro	46.7	36.0	27.3	+19.3 [10.0, 29.3]	+38	-36
OLMo-3-7B	GSM8K	86.7	82.7	79.3	+7.3 [1.3, 14.7]	+8	-6
	MMLU-Pro	39.3	40.7	22.0	+17.3 [10.0, 25.3]	+21	-26
Llama-3-8B	GSM8K	75.3	74.7	71.3	+4.0 [-3.3, 11.3]	+5	-4
	MMLU-Pro	44.0	42.7	27.3	+16.7 [8.7, 24.7]	+29	-32
Mistral-7B	GSM8K	31.3	28.7	32.0	-0.7 [-9.3, 8.0]	+1	-1
	MMLU-Pro	29.3	34.7	20.0	+9.3 [0.7, 18.0]	+9	-8
Qwen2.5-7B	GSM8K	92.7	87.3	88.7	+4.0 [0.7, 8.0]	+1	-4
	MMLU-Pro	65.3	66.0	58.0	+7.3 [-1.3, 16.0]	+7	-10
Gemma-4-E4B	GSM8K	76.0	73.3	73.3	+2.7 [-1.3, 6.7]	+3	+5
	MMLU-Pro	63.3	55.3	58.7	+4.7 [-0.0, 10.0]	+6	-6
Gemma-3-12B	GSM8K	92.0	91.3	88.7	+3.3 [-1.3, 8.7]	+2	-2
	MMLU-Pro	63.3	64.7	60.0	+3.3 [-4.0, 10.7]	+11	-11
Qwen3-8B	GSM8K	92.7	94.0	94.7	-2.0 [-5.3, 1.3]	+1	+2
	MMLU-Pro	60.7	60.0	58.0	+2.7 [-4.7, 9.3]	+6	-4
Qwen3-1.7B	GSM8K	79.3	75.3	74.0	+5.3 [-2.0, 13.3]	+3	+5
	MMLU-Pro	48.0	47.3	47.3	+0.7 [-7.3, 8.7]	+6	-5
Qwen3-4B	GSM8K	96.0	96.7	93.3	+2.7 [-1.3, 6.7]	+0	+0
	MMLU-Pro	64.0	60.7	64.7	-0.7 [-6.0, 4.7]	+2	-1

Table 7: Plain-temperature accuracy drop $T=0.7 \rightarrow 1.3$ (pp, point estimates) at each quantization level of the main grid (150 generations per cell).

Model	Task	Q8_0	Q6_K	Q4_K_M	Q3_K_M
Llama-3.1-8B	GSM8K	+30.0	+32.0	+36.0	+38.0
	MMLU-Pro	+38.0	+46.7	+38.0	+40.7
Mistral-7B	GSM8K	-0.7	-0.7	+3.3	+12.0
	MMLU-Pro	+9.3	+5.3	+10.7	+6.7
Qwen2.5-7B	GSM8K	+4.0	+2.7	+2.7	+6.7
	MMLU-Pro	+7.3	+6.0	+4.0	+2.0
Gemma-3-12B	GSM8K	+3.3	+2.7	-0.0	+2.0
	MMLU-Pro	+3.3	+5.3	+2.7	+3.3
Qwen3-8B	GSM8K	-2.0	+1.3	+2.0	+0.7
	MMLU-Pro	+2.7	-0.7	+0.0	+2.0
Qwen3-4B	GSM8K	+2.7	+0.7	+6.0	+2.7
	MMLU-Pro	-0.7	+0.0	+2.0	+4.0
Qwen3-1.7B	GSM8K	+5.3	-6.0	-0.7	+0.0
	MMLU-Pro	+0.7	+6.0	-8.0	+4.0

Table 8: Stratified MMLU-Pro subset against the fixed business subset: plain-temperature accuracy at Q8_0 and the drop with its interval.

Model	Subset	acc. 0.7	acc. 1.3	drop [95% CI]
Llama-3.1-8B	business (head)	48.0	10.0	+38.0 [28.0, 48.0]
	stratified	44.7	1.3	+43.3 [31.3, 55.3]
Qwen2.5-7B	business (head)	65.3	58.0	+7.3 [-1.3, 16.0]
	stratified	54.7	48.7	+6.0 [-2.0, 14.7]
Gemma-3-12B	business (head)	63.3	60.0	+3.3 [-4.0, 10.7]
	stratified	64.0	56.7	+7.3 [2.0, 12.7]

Table 9: Plain-temperature accuracy drop at each ladder temperature against the model’s own $T=0.7$ accuracy in the main grid (pp, 95% item-clustered intervals; Q8_0).

Task	T	Llama-3.1-8B	Qwen2.5-7B	Gemma-3-12B
GSM8K	1.3	+32.7 [24.7, 41.3]	+4.0 [-0.7, 9.3]	+2.7 [-2.0, 8.0]
	1.5	+68.0 [58.7, 76.7]	+24.0 [15.3, 32.7]	+14.7 [8.7, 22.0]
	1.7	+76.0 [66.0, 85.3]	+70.0 [61.3, 77.3]	+42.7 [34.7, 50.7]
	2.0	+77.3 [67.3, 86.7]	+90.7 [82.0, 97.3]	+84.0 [76.0, 90.7]
MMLU-Pro	1.3	+44.7 [34.6, 55.3]	+8.0 [0.0, 17.3]	+3.3 [-2.7, 10.0]
	1.5	+46.0 [35.3, 56.7]	+40.0 [28.6, 51.3]	+39.3 [29.3, 50.0]
	1.7	+48.0 [36.7, 59.3]	+61.3 [49.3, 73.3]	+57.3 [46.0, 69.3]
	2.0	+48.0 [36.7, 59.3]	+65.3 [53.3, 77.3]	+63.3 [51.3, 75.3]

Table 10: GSM8K accuracy (%) by configuration and temperature (main-grid models pooled over quantization levels; the three flagged models at Q8_0).

Model	Configuration	0.7	1.0	1.3	Model	Configuration	0.7	1.0	1.3
Llama-3.1-8B	greedy	78.0	–	–	Qwen3-4B	greedy	95.2	–	–
	temperature	78.2	74.8	44.2		temperature	95.2	94.0	93.1
	top- p	81.5	78.0	63.8		top- p	95.5	95.5	93.5
	top- k	79.8	75.2	58.7		top- k	94.0	94.5	94.1
	min- p	81.2	78.0	72.0		min- p	95.1	94.7	92.9
	top- $n\sigma$	78.8	80.7	80.8		top- $n\sigma$	94.8	93.9	93.9
	top- p (temp. last)	80.8	78.7	73.0		top- p (temp. last)	94.8	95.2	94.3
	min- p (temp. last)	78.7	81.3	76.2		min- p (temp. last)	94.5	93.9	93.9
Mistral-7B	greedy	31.0	–	–	Qwen3-1.7B	greedy	75.5	–	–
	temperature	32.7	30.0	29.2		temperature	73.3	73.7	73.7
	top- p	28.2	31.0	34.2		top- p	73.8	75.0	72.7
	top- k	31.8	31.7	29.8		top- k	72.8	72.7	73.3
	min- p	32.3	28.3	28.8		min- p	72.2	71.7	74.0
	top- $n\sigma$	29.5	30.7	30.2		top- $n\sigma$	74.3	73.5	72.3
	top- p (temp. last)	29.5	31.2	29.5		top- p (temp. last)	73.0	72.0	71.3
	min- p (temp. last)	29.5	32.0	30.5		min- p (temp. last)	74.7	73.3	74.5
Qwen2.5-7B	greedy	93.5	–	–	Hermes-3-8B	greedy	82.0	–	–
	temperature	91.5	91.7	87.5		temperature	71.3	77.3	60.0
	top- p	92.2	91.2	90.3		top- p	74.7	80.0	71.3
	top- k	90.2	88.5	88.7		top- k	74.0	72.7	67.3
	min- p	90.8	90.5	91.0		min- p	75.3	76.0	71.3
	top- $n\sigma$	92.7	91.2	90.7		top- $n\sigma$	70.7	74.7	77.3
	top- p (temp. last)	90.8	91.3	90.3		top- p (temp. last)	72.0	73.3	77.3
	min- p (temp. last)	92.0	90.7	91.3		min- p (temp. last)	75.3	77.3	74.7
Gemma-3-12B	greedy	92.0	–	–	Qwen3.5-9B	greedy	78.0	–	–
	temperature	91.0	90.2	89.0		temperature	81.3	80.0	74.0
	top- p	91.5	90.3	90.7		top- p	80.7	80.7	80.0
	top- k	91.8	91.7	90.7		top- k	81.3	80.7	77.3
	min- p	91.5	91.3	92.3		min- p	82.7	82.7	83.3
	top- $n\sigma$	91.3	91.5	91.0		top- $n\sigma$	78.7	84.7	81.3
	top- p (temp. last)	90.2	91.8	91.5		top- p (temp. last)	82.7	80.7	82.0
	min- p (temp. last)	91.2	90.8	90.3		min- p (temp. last)	82.0	78.0	79.3
Qwen3-8B	greedy	94.5	–	–	OLMo-3-7B	greedy	90.0	–	–
	temperature	95.0	94.2	94.5		temperature	88.7	81.3	76.0
	top- p	94.0	94.3	93.7		top- p	84.7	88.7	82.0
	top- k	94.8	94.8	93.5		top- k	84.7	85.3	78.7
	min- p	94.8	95.3	94.7		min- p	84.7	83.3	78.0
	top- $n\sigma$	96.3	94.2	94.3		top- $n\sigma$	86.7	84.0	86.0
	top- p (temp. last)	93.8	94.0	94.0		top- p (temp. last)	84.0	85.3	81.3
	min- p (temp. last)	96.2	94.7	95.3		min- p (temp. last)	83.3	84.7	86.0

Table 11: MMLU-Pro accuracy (%) by configuration and temperature (main-grid models pooled over quantization levels; the three flagged models at Q8_0).

Model	Configuration	0.7	1.0	1.3	Model	Configuration	0.7	1.0	1.3
Llama-3.1-8B	greedy	54.0	—	—	Qwen3-4B	greedy	61.6	—	—
	temperature	47.8	39.7	7.0		temperature	61.9	62.5	61.2
	top- p	51.3	46.0	16.0		top- p	62.8	61.3	62.8
	top- k	51.2	43.3	26.8		top- k	63.5	61.1	61.2
	min- p	51.2	50.8	39.5		min- p	60.9	62.7	62.9
	top- $n\sigma$	51.0	50.5	48.8		top- $n\sigma$	62.4	62.4	61.5
	top- p (temp. last)	50.3	46.7	37.5		top- p (temp. last)	62.9	62.7	61.7
	min- p (temp. last)	51.2	50.2	43.5		min- p (temp. last)	61.3	63.6	61.5
Mistral-7B	greedy	35.5	—	—	Qwen3-1.7B	greedy	45.5	—	—
	temperature	30.5	30.2	22.5		temperature	46.0	45.7	45.3
	top- p	30.2	29.5	25.3		top- p	47.2	47.0	47.2
	top- k	32.5	32.3	28.3		top- k	47.5	46.5	45.0
	min- p	33.8	30.7	31.0		min- p	47.0	47.7	46.2
	top- $n\sigma$	32.3	31.5	31.5		top- $n\sigma$	46.8	47.2	46.2
	top- p (temp. last)	29.2	30.2	32.5		top- p (temp. last)	47.7	46.8	43.5
	min- p (temp. last)	33.2	33.3	28.7		min- p (temp. last)	45.7	46.2	49.0
Qwen2.5-7B	greedy	64.0	—	—	Hermes-3-8B	greedy	52.0	—	—
	temperature	62.5	61.3	57.7		temperature	42.7	42.7	26.0
	top- p	65.5	62.8	62.7		top- p	41.3	47.3	38.7
	top- k	62.5	61.7	57.0		top- k	42.7	42.0	34.7
	min- p	61.3	63.5	59.7		min- p	44.0	42.0	46.0
	top- $n\sigma$	62.0	63.5	61.0		top- $n\sigma$	44.7	50.7	38.7
	top- p (temp. last)	63.7	62.7	61.3		top- p (temp. last)	54.7	46.7	43.3
	min- p (temp. last)	64.2	63.0	62.3		min- p (temp. last)	44.0	44.7	44.0
Gemma-3-12B	greedy	61.0	—	—	Qwen3.5-9B	greedy	64.0	—	—
	temperature	63.7	66.7	60.0		temperature	60.0	58.7	38.0
	top- p	65.7	66.7	63.8		top- p	62.0	60.7	53.3
	top- k	64.3	62.8	63.7		top- k	59.3	56.7	53.3
	min- p	65.3	67.0	65.5		min- p	61.3	56.0	55.3
	top- $n\sigma$	65.2	65.8	65.8		top- $n\sigma$	58.0	61.3	59.3
	top- p (temp. last)	65.8	65.0	65.2		top- p (temp. last)	56.7	60.7	54.0
	min- p (temp. last)	65.3	63.7	63.8		min- p (temp. last)	60.0	57.3	57.3
Qwen3-8B	greedy	60.5	—	—	OLMo-3-7B	greedy	46.0	—	—
	temperature	60.5	59.0	59.5		temperature	42.0	41.3	26.7
	top- p	60.5	61.3	60.5		top- p	43.3	39.3	36.7
	top- k	60.2	61.3	57.0		top- k	40.0	40.7	37.3
	min- p	61.0	59.2	59.7		min- p	43.3	40.7	42.7
	top- $n\sigma$	60.0	58.8	58.7		top- $n\sigma$	41.3	42.7	40.0
	top- p (temp. last)	60.7	59.5	60.5		top- p (temp. last)	43.3	44.0	44.7
	min- p (temp. last)	61.3	60.7	59.3		min- p (temp. last)	42.0	44.7	40.0

Table 12: GSM8K: paired accuracy difference of each configuration against plain temperature on the same items (pp, 95% item-clustered bootstrap intervals; main-grid models pooled over quantization, flagged models at Q8_0). Bold: interval above zero.

Model	T	top- p	top- k	min- p	top- $n\sigma$	top- p (temp. last)	min- p (temp. last)
Llama-3.1-8B	0.7	+ 3.3 [0.3, 6.8]	+1.7 [-1.0, 4.5]	+3.0 [-0.8, 7.0]	+0.7 [-2.7, 4.0]	+2.7 [-0.5, 6.3]	+0.5 [-2.3, 3.3]
	1.0	+3.2 [-0.3, 6.8]	+0.3 [-3.5, 4.5]	+3.2 [-0.8, 7.3]	+ 5.8 [2.2, 9.5]	+ 3.8 [0.2, 8.0]	+ 6.5 [2.8, 10.5]
	1.3	+ 19.7 [15.0, 24.3]	+ 14.5 [10.0, 18.8]	+ 27.8 [22.7, 33.2]	+ 36.7 [30.3, 43.2]	+ 28.8 [23.7, 34.2]	+ 32.0 [26.0, 38.3]
Mistral-7B	0.7	-4.5 [-7.7, -1.5]	-0.8 [-5.8, 3.7]	-0.3 [-3.8, 3.3]	-3.2 [-7.7, 1.0]	-3.2 [-8.0, 1.5]	-3.2 [-7.2, 0.7]
	1.0	+1.0 [-2.5, 4.7]	+1.7 [-1.5, 4.7]	-1.7 [-4.8, 1.7]	+0.7 [-2.8, 4.2]	+1.2 [-3.0, 5.5]	+2.0 [-1.8, 6.0]
	1.3	+ 5.0 [0.7, 9.2]	+0.7 [-4.3, 5.5]	-0.3 [-4.0, 3.5]	+1.0 [-2.8, 5.0]	+0.3 [-4.8, 5.0]	+1.3 [-2.3, 5.0]
Qwen2.5-7B	0.7	+0.7 [-1.3, 2.8]	-1.3 [-3.8, 0.7]	-0.7 [-2.0, 0.7]	+1.2 [-0.7, 3.2]	-0.7 [-2.8, 1.3]	+0.5 [-1.0, 2.2]
	1.0	-0.5 [-3.0, 1.7]	-3.2 [-6.0, -0.5]	-1.2 [-3.5, 1.2]	-0.5 [-2.8, 1.5]	-0.3 [-3.3, 2.5]	-1.0 [-3.7, 1.5]
	1.3	+ 2.8 [0.7, 5.2]	+1.2 [-0.7, 3.3]	+ 3.5 [0.8, 6.5]	+ 3.2 [1.2, 5.5]	+ 2.8 [0.5, 5.5]	+ 3.8 [1.5, 6.7]
Gemma-3-12B	0.7	+0.5 [-1.3, 2.5]	+0.8 [-1.0, 3.0]	+0.5 [-1.3, 2.5]	+0.3 [-1.5, 2.5]	-0.8 [-4.0, 2.2]	+0.2 [-2.0, 2.5]
	1.0	+0.2 [-3.0, 3.0]	+1.5 [-0.5, 3.7]	+1.2 [-1.2, 3.7]	+1.3 [-1.0, 3.7]	+1.7 [-1.0, 4.7]	+0.7 [-1.5, 2.8]
	1.3	+1.7 [-0.2, 3.8]	+1.7 [-0.7, 4.2]	+ 3.3 [0.5, 6.7]	+2.0 [-0.0, 4.5]	+ 2.5 [0.3, 4.8]	+1.3 [-0.8, 3.5]
Qwen3-8B	0.7	-1.0 [-3.3, 0.8]	-0.2 [-1.7, 1.0]	-0.2 [-1.7, 1.2]	+1.3 [-0.8, 3.5]	-1.2 [-3.5, 0.8]	+1.2 [-0.8, 3.5]
	1.0	+0.2 [-1.7, 2.0]	+0.7 [-2.0, 3.3]	+1.2 [-1.0, 3.3]	-0.0 [-1.8, 1.7]	-0.2 [-2.0, 2.0]	+0.5 [-1.8, 2.8]
	1.3	-0.8 [-2.5, 0.7]	-1.0 [-2.7, 0.5]	+0.2 [-1.8, 2.3]	-0.2 [-1.7, 1.7]	-0.5 [-2.2, 1.0]	+0.8 [-0.3, 2.2]
Qwen3-4B	0.7	+0.3 [-0.9, 1.7]	-1.2 [-2.9, 0.8]	-0.1 [-1.3, 1.3]	-0.4 [-1.7, 1.1]	-0.4 [-2.0, 1.6]	-0.7 [-2.5, 1.6]
	1.0	+1.5 [0.0, 3.1]	+0.5 [-0.5, 1.7]	+0.7 [-1.1, 2.7]	-0.1 [-1.5, 1.1]	+1.2 [0.0, 2.4]	-0.1 [-1.6, 1.3]
	1.3	+0.4 [-1.2, 2.0]	+1.1 [-0.7, 2.8]	-0.1 [-1.9, 1.6]	+0.8 [-1.1, 2.7]	+1.2 [-0.5, 3.1]	+0.8 [-0.8, 2.5]
Qwen3-1.7B	0.7	+0.5 [-2.8, 3.8]	-0.5 [-4.2, 2.8]	-1.2 [-4.7, 2.2]	+1.0 [-2.2, 4.3]	-0.3 [-3.0, 2.2]	+1.3 [-1.3, 4.3]
	1.0	+1.3 [-2.0, 4.5]	-1.0 [-4.0, 2.0]	-2.0 [-4.3, -0.0]	-0.2 [-2.7, 2.3]	-1.7 [-4.0, 0.5]	-0.3 [-3.2, 2.7]
	1.3	-1.0 [-4.5, 2.3]	-0.3 [-3.5, 2.7]	+0.3 [-2.8, 3.5]	-1.3 [-4.2, 1.2]	-2.3 [-6.0, 1.2]	+0.8 [-2.2, 3.7]
Hermes-3-8B	0.7	+3.3 [-3.3, 10.0]	+2.7 [-3.3, 8.7]	+4.0 [-2.0, 10.0]	-0.7 [-6.0, 4.7]	+0.7 [-6.0, 7.3]	+4.0 [-2.7, 10.0]
	1.0	+2.7 [-3.3, 8.0]	-4.7 [-10.7, 1.3]	-1.3 [-6.7, 4.0]	-2.7 [-9.3, 3.3]	-4.0 [-10.0, 2.0]	+0.0 [-7.3, 7.3]
	1.3	+ 11.3 [2.7, 19.3]	+7.3 [-0.7, 14.7]	+ 11.3 [3.3, 19.3]	+ 17.3 [10.7, 24.0]	+ 17.3 [8.7, 25.3]	+ 14.7 [7.3, 22.0]
Qwen3.5-9B	0.7	-0.7 [-4.7, 3.3]	-0.0 [-5.3, 5.3]	+1.3 [-3.3, 5.3]	-2.7 [-8.0, 2.0]	+1.3 [-3.3, 5.3]	+0.7 [-2.7, 4.7]
	1.0	+0.7 [-5.3, 6.7]	+0.7 [-6.0, 7.3]	+2.7 [-1.3, 7.3]	+4.7 [-1.3, 11.3]	+0.7 [-4.7, 6.7]	-2.0 [-6.0, 1.3]
	1.3	+ 6.0 [1.3, 11.3]	+3.3 [-4.0, 10.7]	+ 9.3 [0.7, 18.0]	+ 7.3 [1.3, 13.3]	+ 8.0 [2.0, 14.7]	+5.3 [-1.3, 12.0]
OLMo-3-7B	0.7	-4.0 [-8.7, 0.7]	-4.0 [-9.3, 0.0]	-4.0 [-9.3, 0.7]	-2.0 [-5.3, 1.3]	-4.7 [-10.7, 0.7]	-5.3 [-10.7, -1.3]
	1.0	+ 7.3 [3.3, 12.0]	+4.0 [0.0, 8.7]	+2.0 [-2.7, 6.7]	+2.7 [-3.3, 8.7]	+4.0 [0.0, 8.7]	+3.3 [-0.7, 8.0]
	1.3	+ 6.0 [1.3, 10.7]	+2.7 [-3.3, 9.3]	+2.0 [-4.0, 9.3]	+ 10.0 [3.3, 17.3]	+5.3 [-0.7, 11.3]	+ 10.0 [3.3, 17.3]

Table 13: MMLU-Pro: paired accuracy difference of each configuration against plain temperature on the same items, as in Table 12.

Model	T	top- p	top- k	min- p	top- $n\sigma$	top- p (temp. last)	min- p (temp. last)
Llama-3.1-8B	0.7	+ 3.5 [0.3, 7.0]	+3.3 [-0.5, 7.5]	+3.3 [-0.7, 8.0]	+3.2 [-0.2, 6.7]	+2.5 [-0.7, 6.2]	+ 3.3 [0.3, 6.7]
	1.0	+ 6.3 [1.8, 11.3]	+3.7 [-1.3, 8.7]	+ 11.2 [6.7, 16.0]	+ 10.8 [6.0, 16.2]	+ 7.0 [3.0, 11.2]	+ 10.5 [6.3, 14.7]
	1.3	+ 9.0 [5.2, 13.3]	+ 19.8 [13.8, 26.7]	+ 32.5 [24.7, 41.0]	+ 41.8 [32.3, 52.2]	+ 30.5 [23.2, 38.7]	+ 36.5 [28.0, 45.5]
Mistral-7B	0.7	-0.3 [-4.5, 4.0]	+2.0 [-2.3, 6.5]	+3.3 [-0.5, 7.0]	+1.8 [-2.2, 6.0]	-1.3 [-4.8, 2.2]	+2.7 [-1.7, 7.3]
	1.0	-0.7 [-4.5, 3.0]	+2.2 [-1.2, 5.5]	+0.5 [-2.8, 3.8]	+1.3 [-3.2, 5.8]	-0.0 [-4.0, 4.5]	+3.2 [-0.8, 7.3]
	1.3	+2.8 [-1.0, 7.0]	+ 5.8 [2.7, 9.3]	+ 8.5 [4.5, 12.5]	+ 9.0 [4.8, 13.5]	+ 10.0 [5.3, 15.3]	+ 6.2 [2.0, 10.3]
Qwen2.5-7B	0.7	+ 3.0 [0.5, 5.8]	+0.0 [-2.7, 2.7]	-1.2 [-4.5, 2.2]	-0.5 [-3.7, 2.5]	+1.2 [-1.2, 3.7]	+1.7 [-1.2, 4.7]
	1.0	+1.5 [-1.8, 5.0]	+0.3 [-2.0, 2.7]	+2.2 [-0.8, 5.3]	+2.2 [-0.8, 5.3]	+1.3 [-1.0, 3.8]	+1.7 [-1.2, 4.8]
	1.3	+ 5.0 [1.7, 8.3]	-0.7 [-3.8, 2.3]	+2.0 [-1.7, 6.5]	+3.3 [-0.2, 7.0]	+ 3.7 [0.2, 7.2]	+ 4.7 [1.0, 8.8]
Gemma-3-12B	0.7	+2.0 [-0.3, 4.5]	+0.7 [-2.0, 3.5]	+1.7 [-0.8, 4.3]	+1.5 [-1.2, 4.3]	+2.2 [-1.0, 5.7]	+1.7 [-2.0, 5.7]
	1.0	-0.0 [-2.8, 2.7]	-3.8 [-7.2, -1.0]	+0.3 [-2.2, 2.7]	-0.8 [-3.0, 1.3]	-1.7 [-4.7, 1.3]	-3.0 [-6.2, -0.2]
	1.3	+ 3.8 [0.2, 8.3]	+ 3.7 [0.3, 7.5]	+ 5.5 [2.7, 8.8]	+ 5.8 [2.3, 9.8]	+ 5.2 [2.3, 8.3]	+ 3.8 [0.5, 7.5]
Qwen3-8B	0.7	-0.0 [-2.8, 3.0]	-0.3 [-2.7, 2.0]	+0.5 [-2.3, 3.7]	-0.5 [-2.8, 1.8]	+0.2 [-2.7, 2.8]	+0.8 [-2.2, 3.8]
	1.0	+2.3 [-0.5, 5.2]	+2.3 [-0.7, 5.3]	+0.2 [-3.2, 3.3]	-0.2 [-3.2, 2.7]	+0.5 [-2.3, 3.2]	+1.7 [-1.2, 4.5]
	1.3	+1.0 [-1.8, 3.5]	-2.5 [-5.7, 0.7]	+0.2 [-3.0, 3.5]	-0.8 [-4.2, 2.3]	+1.0 [-1.8, 3.8]	-0.2 [-3.5, 3.0]
Qwen3-4B	0.7	+0.9 [-1.5, 3.1]	+1.6 [-0.5, 3.7]	-0.9 [-3.5, 1.6]	+0.5 [-1.5, 2.5]	+1.1 [-1.1, 3.2]	-0.5 [-3.1, 2.0]
	1.0	-1.2 [-3.2, 0.9]	-1.5 [-3.5, 0.4]	+0.1 [-2.7, 3.2]	-0.1 [-2.7, 2.4]	+0.1 [-2.7, 3.2]	+1.1 [-1.5, 3.7]
	1.3	+1.6 [-1.2, 4.3]	+0.0 [-2.7, 2.5]	+1.7 [-0.7, 4.1]	+0.3 [-2.3, 2.8]	+0.5 [-2.3, 3.2]	+0.3 [-2.1, 2.8]
Qwen3-1.7B	0.7	+1.2 [-2.0, 4.2]	+1.5 [-1.7, 4.7]	+1.0 [-2.5, 4.7]	+0.8 [-2.5, 4.2]	+1.7 [-1.3, 4.8]	-0.3 [-3.7, 2.8]
	1.0	+1.3 [-2.7, 5.0]	+0.8 [-2.2, 4.2]	+2.0 [-1.3, 5.3]	+1.5 [-1.8, 4.8]	+1.2 [-2.3, 4.8]	+0.5 [-3.2, 4.2]
	1.3	+1.8 [-2.8, 7.0]	-0.3 [-4.3, 3.5]	+0.8 [-3.8, 5.5]	+0.8 [-3.5, 5.2]	-1.8 [-6.5, 2.8]	+3.7 [-0.7, 8.7]
Hermes-3-8B	0.7	-1.3 [-8.7, 6.0]	+0.0 [-8.7, 8.0]	+1.3 [-6.0, 8.0]	+2.0 [-4.7, 8.7]	+ 12.0 [4.7, 19.3]	+1.3 [-6.7, 9.3]
	1.0	+4.7 [-2.7, 12.0]	-0.7 [-8.0, 6.7]	-0.7 [-8.7, 7.3]	+ 8.0 [1.3, 14.7]	+4.0 [-1.3, 9.3]	+2.0 [-6.7, 10.7]
	1.3	+ 12.7 [5.3, 20.0]	+ 8.7 [2.0, 16.0]	+ 20.0 [12.0, 29.3]	+ 12.7 [4.0, 22.0]	+ 17.3 [8.7, 26.7]	+ 18.0 [10.0, 26.7]
Qwen3.5-9B	0.7	+2.0 [-1.3, 5.3]	-0.7 [-5.3, 4.0]	+1.3 [0.0, 3.3]	-2.0 [-6.7, 2.0]	-3.3 [-8.0, 0.0]	-0.0 [-4.7, 4.0]
	1.0	+2.0 [-1.3, 5.3]	-2.0 [-6.7, 1.3]	-2.7 [-7.3, 2.0]	+2.7 [-2.0, 7.3]	+2.0 [-4.0, 8.0]	-1.3 [-6.0, 3.3]
	1.3	+ 15.3 [9.3, 22.7]	+ 15.3 [8.0, 24.0]	+ 17.3 [10.0, 26.0]	+ 21.3 [13.3, 30.0]	+ 16.0 [9.3, 23.3]	+ 19.3 [12.0, 28.0]
OLMo-3-7B	0.7	+1.3 [-4.0, 6.7]	-2.0 [-8.7, 4.0]	+1.3 [-4.7, 8.0]	-0.7 [-5.3, 3.3]	+1.3 [-4.0, 6.7]	-0.0 [-6.0, 5.3]
	1.0	-2.0 [-8.7, 4.7]	-0.7 [-8.0, 6.7]	-0.7 [-7.3, 6.0]	+1.3 [-6.0, 8.7]	+2.7 [-4.0, 9.3]	+3.3 [-4.0, 10.7]
	1.3	+ 10.0 [2.7, 17.3]	+ 10.7 [4.0, 17.3]	+ 16.0 [8.0, 24.0]	+ 13.3 [6.0, 22.0]	+ 18.0 [10.7, 26.0]	+ 13.3 [6.6, 20.7]